

COMICCRAFT

DOCUMENT 11: PERFORMANCE TESTING

Step 1: Testing Overview

FIELD	DETAILS
Testing Tool Used	Python time module (manual benchmarking), Browser DevTools Network tab, Postman for API testing
Type of Testing	Load Testing, Response Time Testing, Functional Validation
Target Module / API	POST /generate (full pipeline), POST /generate-comic/json (JSON API), GET /download/{id} (PDF download)
Test Environment	Local development machine (Windows 11, Python 3.12, 16GB RAM, NVIDIA GTX 1650)
Test Date	March 2026

Step 2: Test Scenarios

S.NO	TEST SCENARIO / DESCRIPTION	NO. OF VIRTUAL USERS	DURATION (SEC)	EXPECTED OUTCOME
1	Basic story generation (text only - Gemini Flash + Pro, no image gen)	1	5s	Story outline + narration generated in < 5 seconds
2	Full pipeline (story + 5 panel images via Stable Diffusion/FLUX)	1	30s	Full comic with all 5 images generated in < 30 seconds
3	PDF export from existing comic data (no generation, just compilation)	1	3s	PDF compiled and downloadable in < 3 seconds
4	Mock fallback mode (no API keys - full pipeline with placeholder content)	1	2s	Placeholder comic generated instantly in < 2 seconds

Step 3: Performance Test Results

S.NO	METRIC	TARGET VALUE	ACTUAL VALUE	STATUS	REMARKS
1	Response Time (Avg)	< 2 seconds (text only)	1.2 seconds	■ Pass	Story generation with Gemini (Flash + Pro)
2	Response Time (Max)	< 5 seconds (text only)	4.5 seconds	■ Pass	With complex/long prompts
3	Throughput (Req/sec)	N/A (single user)	N/A	■ Pass	Single-user local testing
4	Error Rate	< 1%	0%	■ Pass	No errors during all test runs
5	CPU Utilization	< 80%	~45%	■ Pass	Efficient text processing

6	Memory Utilization	< 80%	~200MB	■ Pass	Lightweight memory footprint (without image loading)
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Step 4: Observations & Analysis

The application performs well within expected parameters. Text-based generation using Gemini Flash + Pro completes in under 2 seconds on average, well within the 5-second target. Image generation via Stable Diffusion is the primary bottleneck (taking 15-25 seconds for 5 images on a GTX 1650), which is expected for local GPU inference. The mock fallback mode provides near-instant results (under 2 seconds) making it ideal for UI testing and demonstrations without API costs. PDF export is consistently fast at under 2 seconds. Overall, the application meets all performance targets for the MVP scope.